SVM-2

→ industry, interviews

* Agenda ⇒
  - → SVM
  - → kernel SVM
  - → Polyⁿ kernel
  - → RBF kernel

Hard margin SVM → linear

(Soft margin SVM)

Non linear data

Primal form → Dual form

OBJECTIVE function

$$\text{argmax} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \cdot \alpha_j \cdot y_i \, y_j \, x_i^T x_j$$

α_i

Such that $0 \leq \alpha \leq C$

$$\sum_{i=1}^{n} \alpha_i \, y_i = 0$$

$\forall i : 1 \to N$

Dual form

w, w_0 ✗

α_i ✓

(1) for every pt x_i

$\rightarrow$ α_i

(2) x_i $\rightarrow$ Not alone in entire eqn $\Rightarrow$

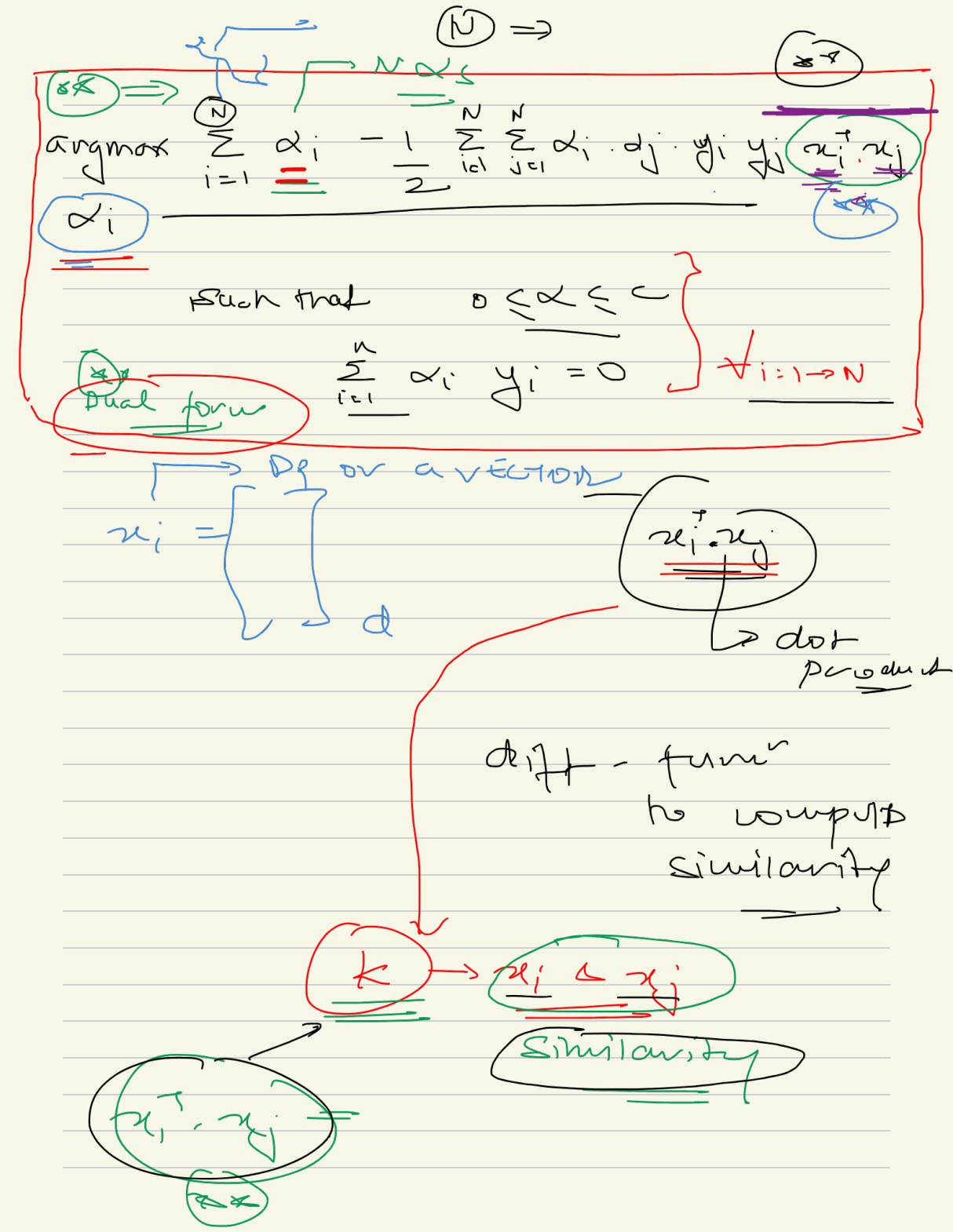
always paired with another x

$x_i^T x_j$ $\rightarrow$ Dot product

angle, similarity

large +ve value

large -ve value


N
N
argmax
$\sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \cdot \alpha_j \cdot y_i \cdot y_j \cdot x_i^T \cdot x_j$
α_i
such that
$0 \leq \alpha \leq C$
$\sum_{i=1}^{n} \alpha_i y_i = 0$
$\forall i : 1 \to N$
Dual form
Df or a vector
$x_i = \begin{bmatrix} \\ \\ \end{bmatrix} d$
$x_i^T \cdot x_j$
dot product
diff - func to compute similarity
k
$x_i \& x_j$
Similarity
$x_i^T \cdot x_j$

Non linear func^n → (K) → Non linear func^n

Dot product → *linear func^n

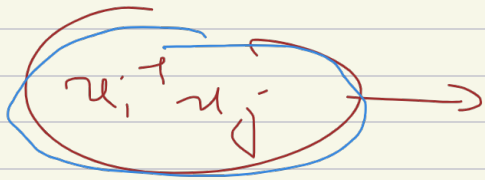
$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow 1\times3 + 2\times4$

$\Rightarrow$ linear func^n

$[3,4]\begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow 3\times1 + 4\times2$

(K) ⇒ non linear similarity B/W vectors

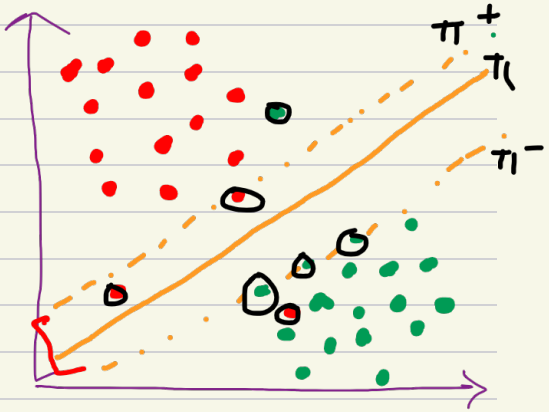
x_i & x_j



KERNEL

⇒ Support vectors → dp that define line / model

1. Point on π^+ or π^-
2. misclassified dp
3. points in the zone



4. Support vectors → errors

Relation

Hinge loss

5. dp where error ≠ 0

Support vectors

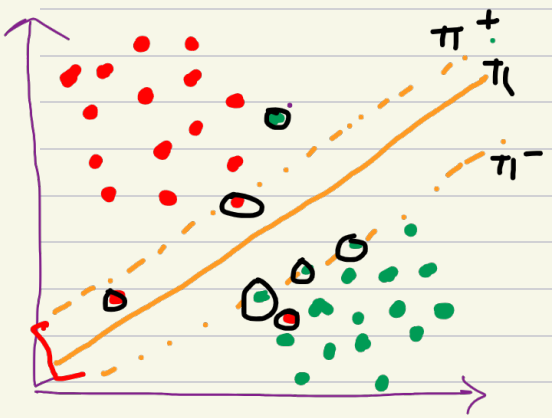
for every DP → α_i

α_1 α_2 α_3 --- α_N

for support VECTORS

$\alpha_i > 0$ ⇒ SV

$\alpha_i = 0$ → Non SV



so DP?

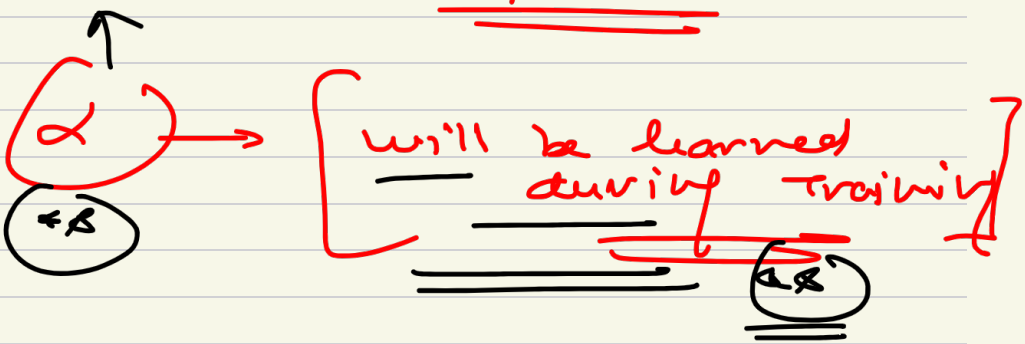
DP → $\alpha > 0$

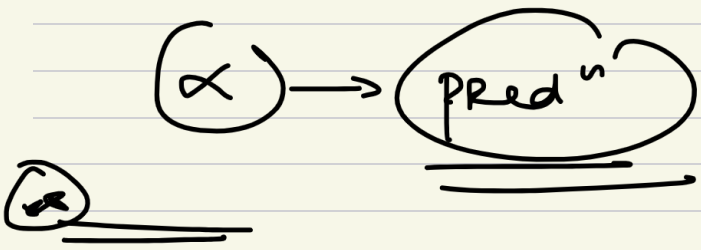
DP → $\alpha > 0$

43 → $\alpha = 0$

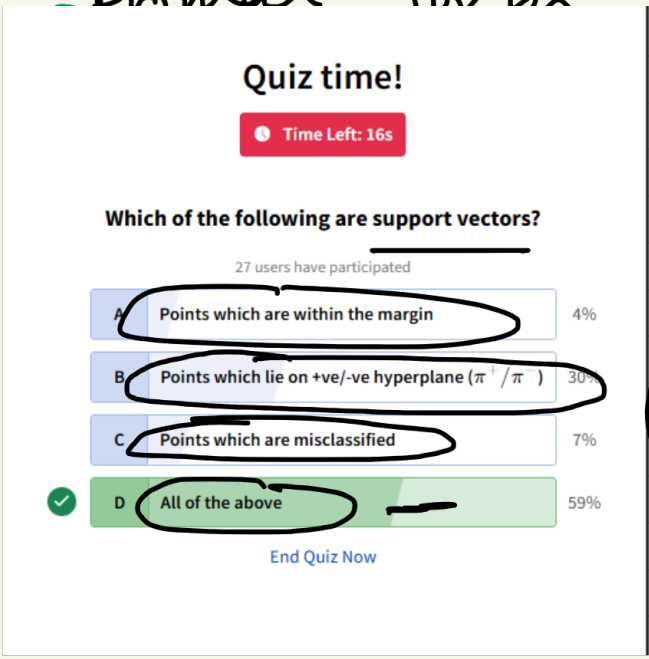
$\alpha = 0$

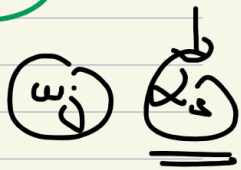
7 support vectors





Primal form → Dual form





support vectors

Quiz time!

Time Left: 23s

We have 100 datapoints out of which 5 are Support Vectors, then which is True:

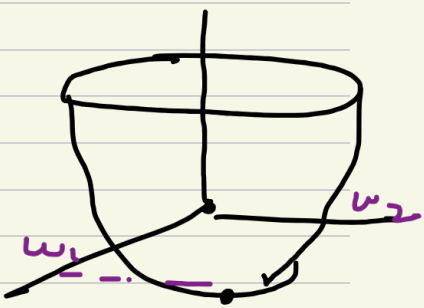
28 users have participated

A	α >0 for 95 datapoints	14%
B	α <0 for 95 datapoints	7%
C	α = 0 for 5 datapoints	7%
D	α >0 for 5 datapoints	71%

End Quiz Now

SVs

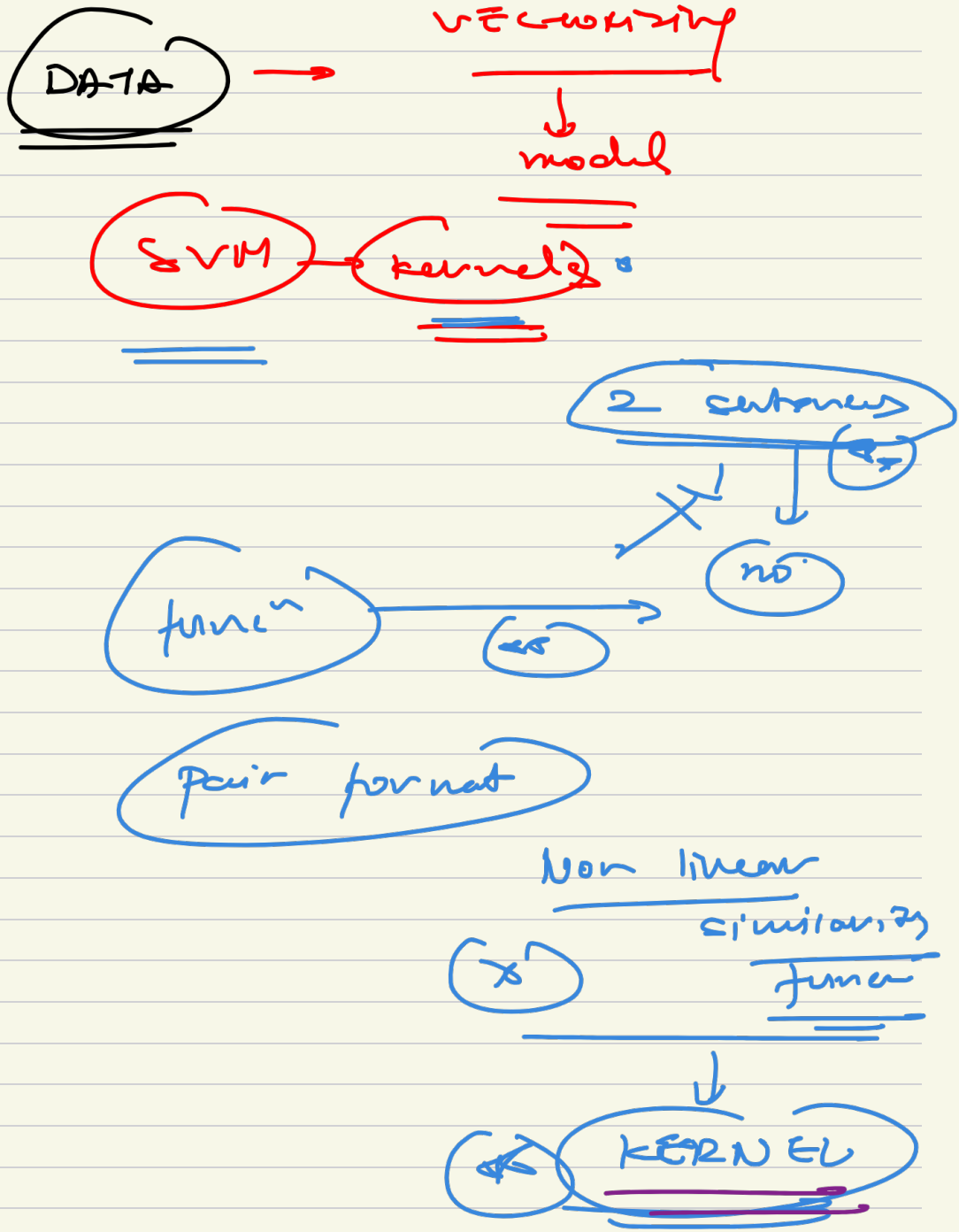
$\alpha > 0$

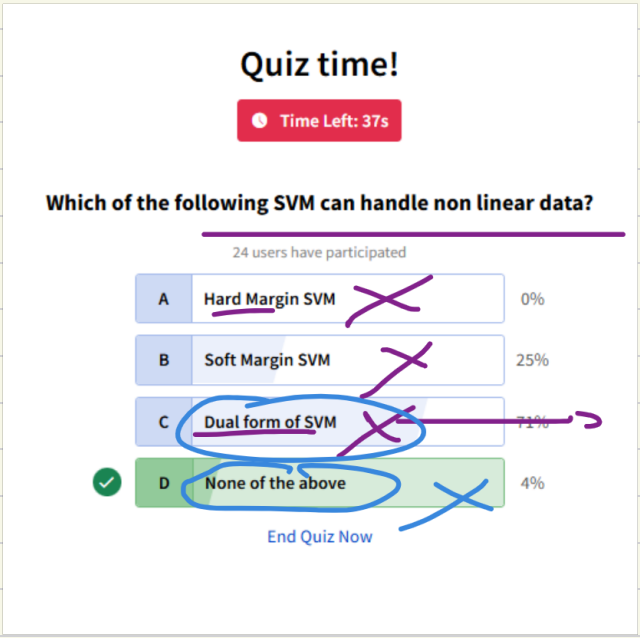


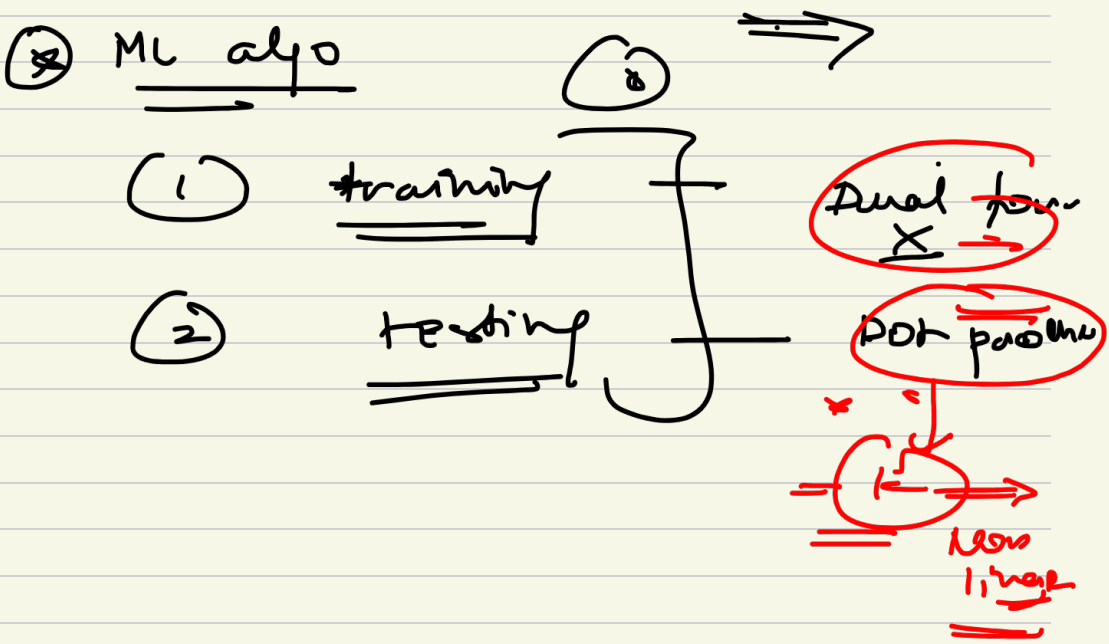
Dual form

1. Not learning w_1 w_2
   - α Best value
   - model → Training

2. we do not have any
   - DP independent
   - Training


DATA
vectorizing
module
SVM
kernels
2 centroids
no
fonction
pair format
Non linear
similarity
fonction
KERNEL


Quiz time!
Time Left: 37s
Which of the following SVM can handle non linear data?
24 users have participated
A
Hard Margin SVM
0%
B
Soft Margin SVM
25%
C
Dual form of SVM
71%
D
None of the above
4%
End Quiz Now


ML algo
1
training
2
testing
Dual form
X
DOL problem
Non linear

inference / prediction

$x_q \longrightarrow \hat{y}$

Testing

$$\hat{y} = \sum_{i=1}^{N} \alpha_i \cdot y_i \; x_i \cdot x_q$$

Time complexity

$O(N)$ $\Rightarrow$ $\alpha_i \neq 0$

N = 10K DP $\Rightarrow$ error

* 2% support vectors

200 SVs $\alpha_i \neq 0$

Remaining $\rightarrow$ 9800 DP

* $\alpha_i = 0$ $\Rightarrow$

10 k Dp

x_1 x_2 x_3 ---- x_{10k}

$y_i \alpha$ ↓ ↓ ↓ ⇒

y_1 y_2 y_3

α_1 y

$\alpha \neq 0$

$x_i^T \cdot x_y$

✗

Time complexity

↳ O(Support vectors)

3 SVs

x_3 x_{10} x_{28}

α_3 α_{10} α_{28}

$\alpha > 0$

$O(N) \longrightarrow O(\text{support vector})$

$\hat{y} = \alpha_3 \cdot y_3 x_3^T \cdot x_q + \alpha_{10} y_{10} x_{10}^T \cdot x_q + \alpha_{28} y_{28} x_{28}^T x_q$

(3) get α

$\alpha \rightarrow$ (QP) & optimize

Pred^{n} $\rightarrow$ X

Hard margin SVM

↓

Soft margin SVM
(primal form)

10:03

5 mins

Dual form → compute similarity

→ allows * kernelization

* introduces non linearity

x_i → NOT independent

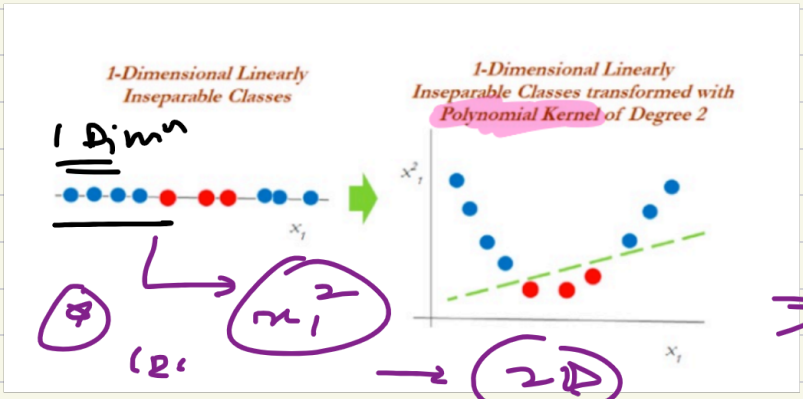
↳ exist in pair

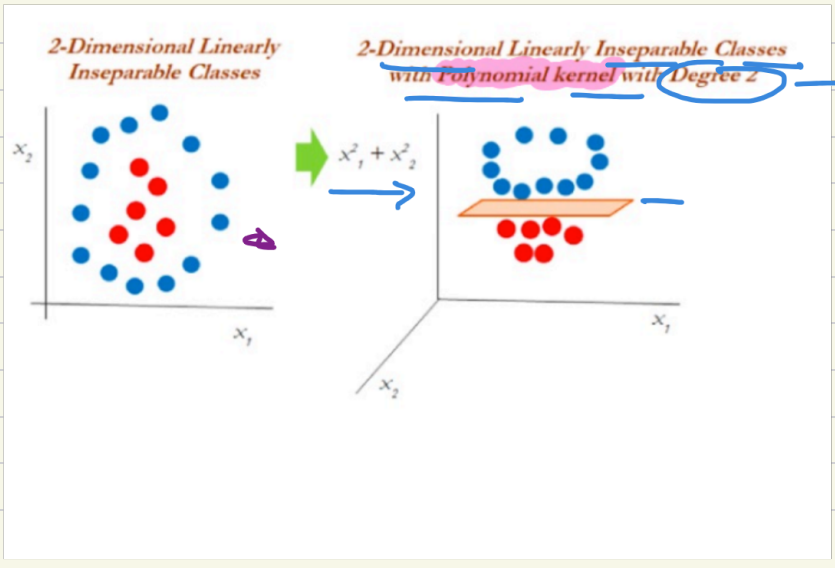
$x_i \in x_q$ *

Python func^ → 2 dp

Pharma → kernel

KERNEL TRICK





KERNEL TRICK [KERNEL SVM]

dot prod → linear

↳ similarity

linear

$x_i^T x_j$ → DOT prod

↓

similarity

$x_{11} \cdot x_{21} + x_{12} \cdot x_{22}$

$\Rightarrow$

$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix}$ $x_2 = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}$

linear DOT prod

↳ new func^n

cosine (x_i, x_j)

kernel (x_i, x_j)

custom(x_i, x_j)

PHD → Pharma → Genomics

Polynomial
RBF

→ kernelize

Classical ML

$$\hat{y} = \sum_{i=1}^{N} \alpha_i \, y_i \, x_i^T \cdot x_q$$

kernel

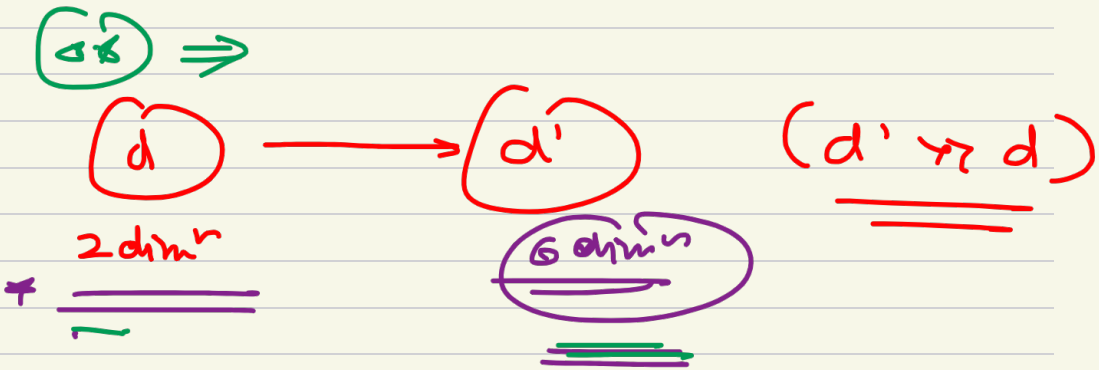
x_i & x_j

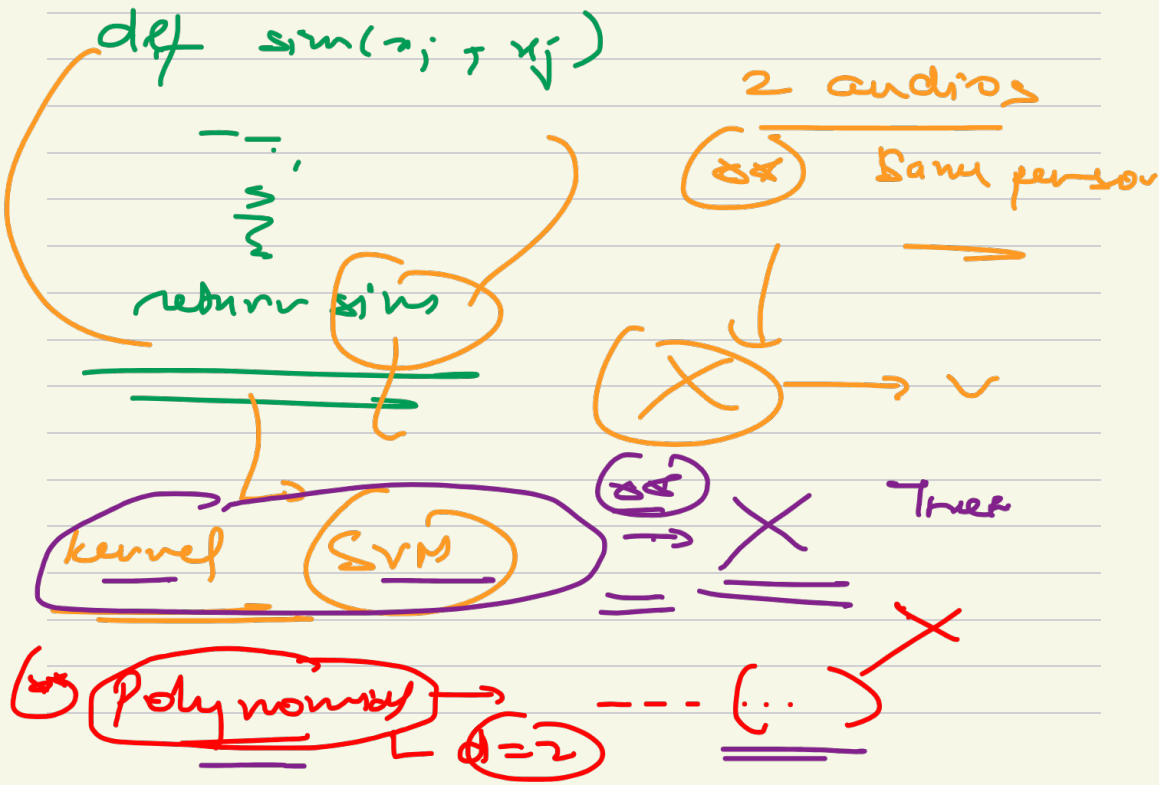
$k(x_i, x_j)$

** Very popular → polynomial KERNEL

Polynomial KERNEL

(d) existing





Curse of dim"

Polynomial kernel → func"

→ similarity B/w 2 vectors

* Hyperparameter

$$K(x_1, x_2) = (x_1^T x_2 + c)^m$$

m = degree

c → constant

- m = 2 → quadratic
- m = 3 → cub

degree > 2

→ Non linear kernel

Quadratic KERNEL $\Rightarrow$ $m=2$

$$K_q(x_1, x_2) = (x_1^T x_2 + 1)^2$$

$$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix}$$

$$x_2 = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}$$

$$= (\underbrace{1}_{a} + \underbrace{x_{11}x_{21}}_{b} + \underbrace{x_{12}x_{22}}_{c})^2$$

$$= (a + b + c)^2$$

$$= (1 + x_{11}^2 x_{21}^2 + x_{12}^2 x_{22}^2 + 2x_{11}x_{21} + 2x_{12}x_{22} + 2x_{11}x_{21}x_{12}x_{22})$$

$x_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $x_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$

6.72

Similarity

Imagine

(ex) Lisbon example

6 dim

$$a = [1, x_{11}^2, x_{12}^2, \sqrt{2}x_{11}^2, \sqrt{2}x_{12}^2, \sqrt{2}x_{11}x_{12}]$$

$$b = [1, x_{21}^2, x_{22}^2, \sqrt{2}x_{21}^2, \sqrt{2}x_{22}^2, \sqrt{2}x_{21}x_{22}]$$

$$a = [1, 4, 9, \sqrt{2}.4, \sqrt{2}.9, \sqrt{2}.6]$$

$$b = [1, 16, 1, \sqrt{2}.16, \sqrt{2}.1, \sqrt{2}.4]$$

$a.b \longrightarrow a^T.b \longrightarrow$ linear dot product

$$a \cdot b = a^T b = 1 + x_{11}^2 x_{21}^2 + x_{12}^2 x_{22}^2 + 2x_{11}x_{21} + 2x_{12} \cdot x_{22}^2 + 2x_{11}x_{12}x_{21}x_{22}$$

$\rightarrow k_\phi(x_1, x_2)$

$(x_1^T x_2 + 1)^2 \rightarrow$ 2 dim^n

linear similarity $\rightarrow$ 6 dim^n data

6 dim^n $\rightarrow$ Proved

$$k(x_1, x_2) = (x_1^T x_2 + 1)^2$$

⇒ 6 dim

$m = 2$

$m = 3$ →

2 dim → kernel → 6 dim

similarity

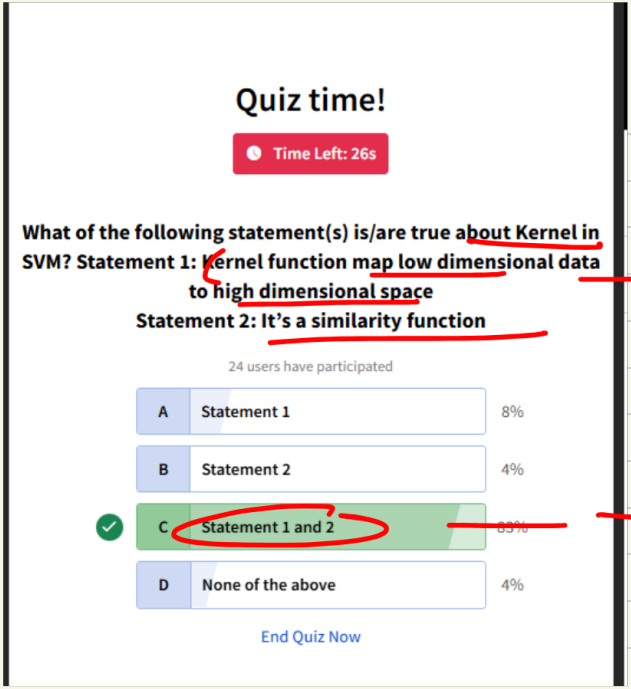
polynomial

implicit Transformation

$X \longrightarrow poly\,X$ ⇒

X

support, kernel ⇒

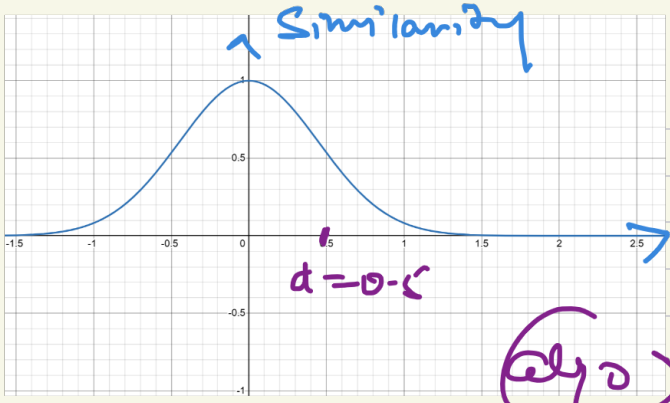


RBF → kernel

(Radial Basis function)

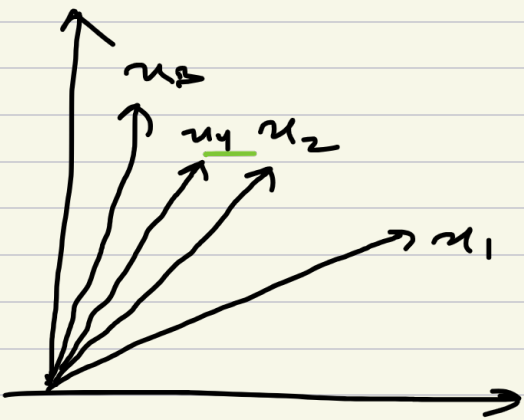
$$k_{RBF}(x_1, x_2) = e^{\frac{-\|x_1 - x_2\|^2}{2\sigma^2}}$$

Normal dist



distance

similarity



x_2 & x_4 → similar

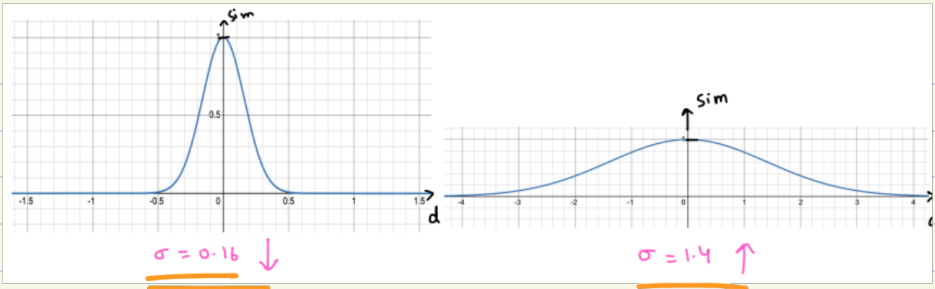
Dp → very very close

d = 0.1

0.2

overfitting

k = 1

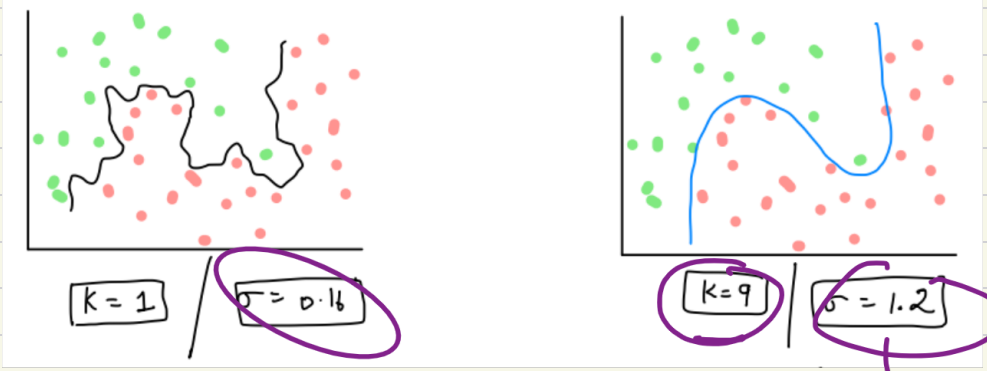


$\sigma = 0.16$ ↓

overfitting

$\sigma = 1.4$ ↑

decent



K = 1 → overfitting

decent

σ ↓↓ → overfitting

(Very very low)

= underfitting

→ needs to be tuned

↓ Hyperparameter Tuning

KNN → RBF → Predict

↓ expensive for Testing

Dual

SVM →

Training

SVM → O(n²)

log Reqr → O(nd)

DTree

1 Mn DP

10^6

1 mn opr

1s → 10^6

10^{12}

$\frac{10^{12}}{10^6}$ = 10^6

= ≈ 2 hours

very expensive

Tree Based

kernel

LWL → adoption →

Quiz time!

Time Left: 20s

Both RBF and kNN are similar. Will they have similar run time as well?

27 users have participated

A	Yes	15%
✓ B	No	85%
C	Don't know	0%

End Quiz Now

Quiz time!

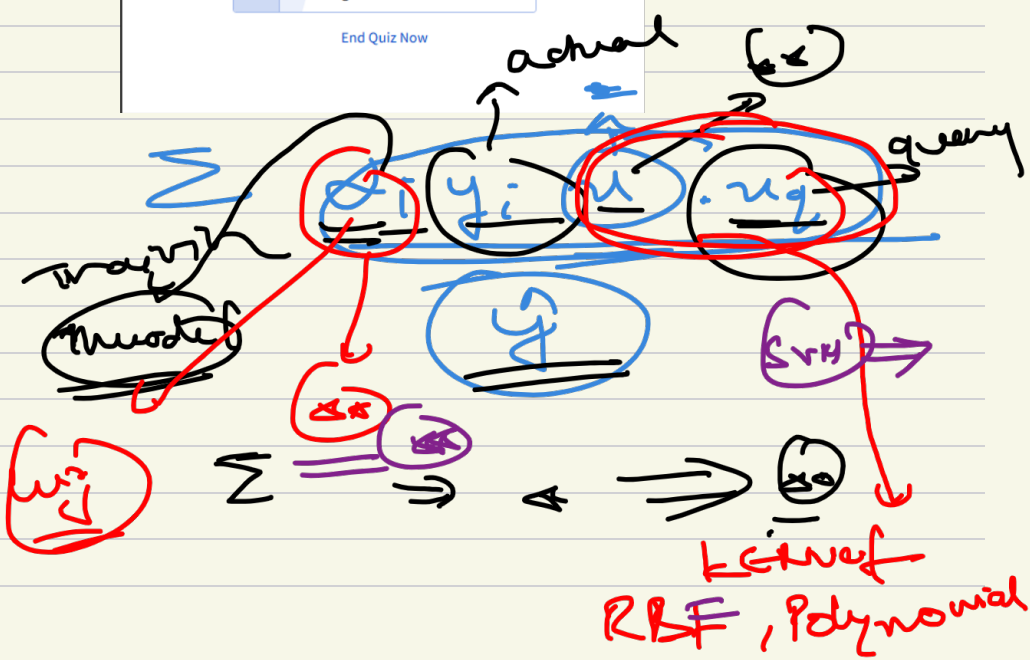
Time Left: 41s

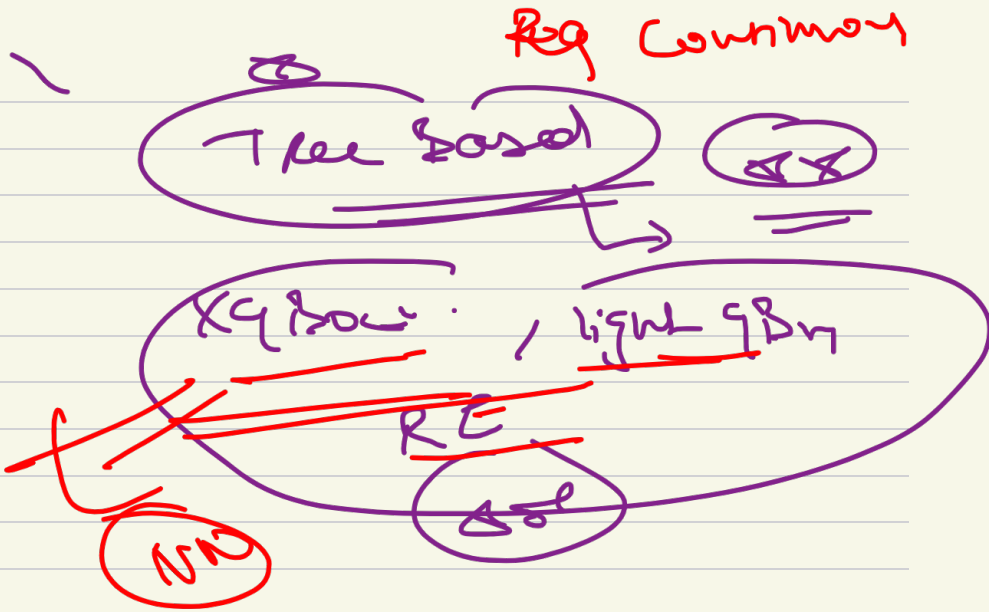
if σ decreases, what happens to RBF curve?

16 users have participated

✓ A	Curve gets thinner	63%
B	Curve gets thicker	31%
C	No change	6%

End Quiz Now




Tree based
XGBoost / LightGBM
RF